

Spatial Filter Bank Based on Probabilistic Common Spatial Pattern in Motor Imagery BCI Systems

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Abstract—Recently, Motor Imagery (MI) based on Brain Computer Interface (BCI) systems have been noticed for neuro-rehabilitation methods. The challenging key is to correct detection of MI tasks. Probabilistic Common Spatial Pattern (P-CSP) is the most recent and efficient method for discriminating two classes of electroencephalogram (EEG) from Motor Imagery (MI) Task. P-CSP resolves the overfitting which is the main issue of spatial filters. The accuracy of true detection tasks is related to some initial values, like the number of sources. In this paper, we generate a feature set by extracted features of each unique filter and select discriminant features by sparse dictionary learning method. Optimal sparse regularization parameter is selected by cross-validation on training data and the maximum accuracy with the corresponding parameter is reported for each subject. The performance of the proposed method is evaluated on publicly available BCI Competition IV dataset 2a to detect two pair classes. Our results have been compared with P-CSP, by sweeping the Best number of source and using automatic selection methods. The results show that sparse features selection from feature set outperforms the existing P-CSP in terms of classification accuracy, and reduces the computational time of selecting the best number of the sources by sparse selection.

Keywords- EEG; Brain-Computer Interface; Probabilistic Common Spatial Pattern; Sparse Feature Selection

I. INTRODUCTION

Brain-Computer Interface (BCI) is the most popular technique in neurological rehabilitation, which creates a communication pathway between the brain and environment without any muscular activity. This kind of communication can help several disabled people by non-muscular brain signal without using peripheral nerves [1]. Motor Imagery (MI) Tasks are prevalent to create the non-muscular brain activities. Electroencephalography (EEG) is a feasible and low-cost brain activities recording method in MI-based BCI systems, which mainly leads to high temporal resolution [2]. EEG is a non-invasive recording method by multiple electrodes placed on the scalp, in order to obtain electrical neural masses activity. These brain states could have significant activities through motor imagery by movement tasks.

As the EEG measurements are the mixture of electrical brain sources activities and other non-brain sources such as physiological and instrumental artifacts, reliable detection of motor imagination is a key challenge in MI-BCI systems. To obtain the local activities from desired brain source, spatial filters are proposed. The fundamental of these filters is to enhance the local activities and to suppress the non-brain sources activities as well as improving the signal to noise ratio (SNR) [3].

Common Spatial Pattern (CSP) is a very efficient method for feature extraction of two conditioned motor imagery EEG data. CSP provides a spatial filter which maximizes the variance of one class and minimizes the variance of another simultaneously [4]. To improve and obtain more acceptable features, several effective methods were used, of which the most important ones are categorized as follows. To combine the spatial and spectral patterns, Common Spatio-Spectral Pattern (CSSP) [5] and Common Sparse Spatio-Spectral Pattern (CSSSP) [5] were proposed. CSSP optimizes a first-order finite impulse response (FIR) temporal filter by adding one-time-delay sample. CSSSP improves the flexibility of CSSP by simultaneously optimizing an arbitrary FIR filter in CSSP. According to several recent researches, the spatial pattern is sensitive to the appropriate selection of frequency bands. Filter Bank CSP (FBCSP) [6] and Sub-Band CSP (SBCSP) [7] are the primary researches on spatial filter for optimized frequency sub-band. FBCSP combined a filter bank framework with CSP and SBCSP optimized a spectral filter within CSP by using multiple sub-band CSP features. Mutual Information-based Best Individual Feature (MIBIF) [8] [9], Welch's t-test, F ratio [10], Fisher score and Laplacian score [11], and Sparse Dictionary Learning [12] are some of the most effective methods used for feature selection in all sub-bands.

Another category of the research on CSP is the stationary contribution. Since EEG could be regarded as a mixture of underlying stationary and nonstationary sources, in Stationary Subspace Analysis (SSA) [13], the observed signal has been modeled as a linear superposition of stationary and nonstationary sources. SSA and CSP have been applied as a kind of preprocessing procedure and stationary sources of feature

extraction, respectively and classifier performance has been significantly improved by the preprocessing of SSA in [14]. The recent method on Regularized CSP [15] is to minimize within-class dissimilarities besides maximizing discrimination between classes' means which is called Kullback-Leibler (KL) CSP. The most recent approach in spatial filtering is probabilistic CSP (P-CSP) with maximum a posteriori (MAP)-based iterative updating algorithm [16]. This method contains some initial value such as number of sources and the noise distribution. The classification accuracy in MAP-CSP depends on the number of sources which needs an automatic method to be detected. [17] Proposed three information based methods, including Akaike Information Criterion (AIC), Minimum Description Length (MDL) and Bayesian Information Criteria (BIC) to estimate the number of sources.

In this paper, we proposed a method based on selecting discriminant features when there are a varying number of sources. Here we generate a feature set by features of each initial value which is the number of largest eigenvalues in signal Eigen decomposition. In practice, the feature obtained by each number of source has not enough discriminants, although we can select best features by selecting best features among feature set.

The rest of this paper is organized as follows: Section II introduces spatial filters and sparse feature selection. In section III, applied dataset and evaluation scheme and the result of performed experiment are explained. Finally, the conclusion of our research is explained in section IV with a discussion of the methods. Fig. 1 illustrates a roadmap of the procedure of proposed algorithm.

II. METHODS

A. Feature Extraction and Spatial Patterns

Common Spatial Pattern (CSP) is an effective method to discriminate two classes of data in feature selection section. As mentioned in part I, CSP is currently the most widespread tool used in motor imagery BCI systems where EEG feature extraction is required. The method is based on diagonalizing of two spatial covariances simultaneously. In this paper, CSP was used for a two-class problem. The spatial filter maps the brain source activity to the electrodes.

Consider EEG samples of two classes, represented by $X \in R^{N \times L \times 2}$, where N is the number of channels and L is the number of samples in trials. Also, let $\hat{R}_1$ and $\hat{R}_2$ be the covariance matrix of each class. One of the basic methods to acquire desired spatial filter as W , is maximizing the following cost function:

$$J(w) \triangleq \frac{w^T X_1 X_1^T w}{w^T X_2 X_2^T w} = \frac{w^T \hat{R}_1 w}{w^T \hat{R}_2 w}, \quad (1)$$

One of the most recent and effective methods in spatial filters is the use of Probabilistic Common Spatial Filter (P-CSP) [16]. The base of PCSP is to present a probability model of the multichannel EEG signals. To estimate this probability model, some subset methods like Maximum A posteriority (MAP) and Variational Bayesian (VB) have been proposed. Suppose $X \in R^{N \times L \times K}$ is the observation signal expresses as a 3-way tensor, where N , L and K are channel, time and condition respectively. Similarly, $Z \in R^{M \times L \times K}$ represents brain source activity which M is number of components. $E_k \in R^{N \times L}$ illustrates the Gaussian noise added to class k . The signal can be modeled as follows:

$$X_k = AZ_k + E_k \quad (2)$$

$$A_{n, \cdot} \sim N(0^T, \Xi), Z_{\cdot, l, k} \sim N(0, \Lambda_k), E_{\cdot, l, k} \sim N(0, \Psi_k)$$

Here $A \in R^{N \times M}$ is a matrix that maps the electrical brain source to the measured signals from the electrodes. Generally, the original EEG signal with fewer independent components can be shown by $M \leq N$. Here A and Z can be estimated by updating MAP estimation to increase the posterior. To resolve some problems like trapping in local optima and unknowing the initial distribution parameters (Λ_k, Ψ_k, Ξ) in practice, [14] proposes MAP-CSP and VB-CSP. While the number of the independent source (M) and noise distribution are known, the MAP-CSP can optimize A and Z estimation and decrease local optima issue. VB-CSP is another probabilistic method which can estimate A and Z when we have no information about the distribution of

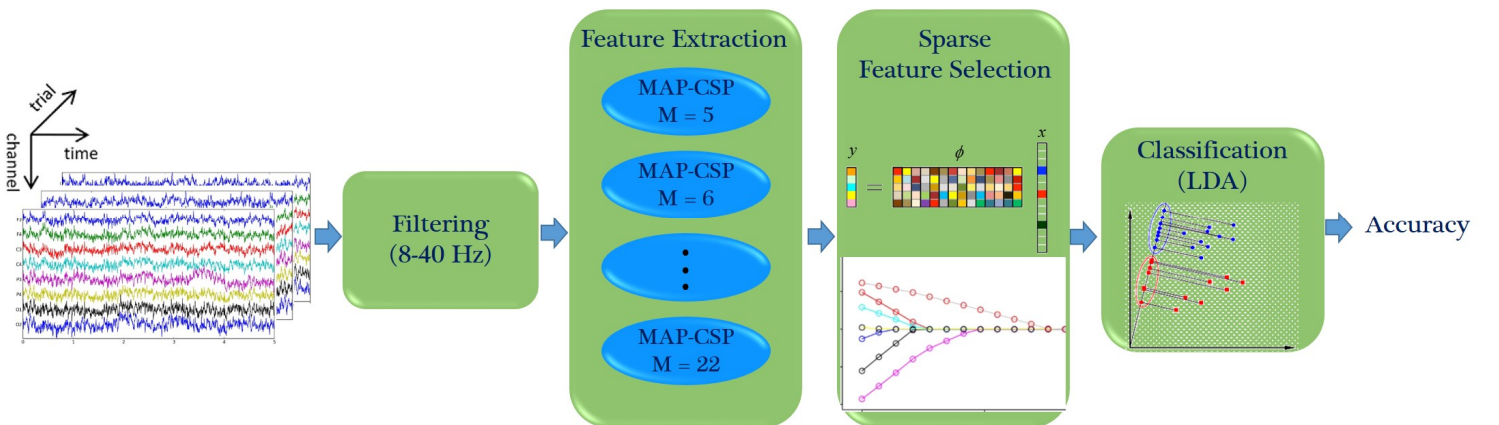


Figure 1. An illustration of the proposed spatial filter bank based on probabilistic common spatial pattern and sparse feature selection

Λ_k, Ψ_k, Ξ . VB-CSP is based on Bayesian inference methods and thus has high computational cost, hence, is not proper for the online algorithm on BCI applications. In the present study, MAP-CSP is used to generate proper spatial patterns.

The MAP which estimates A and Z in (2) can be formulated by the following optimization problem:

$$\min_{A, Z_k} \sum_k \psi_k^{-1} \|X_k - AZ_k\|_2^2, \quad (3)$$

Which $\|\cdot\|_2$ is the ℓ_2 norm. Mentioned equation is equivalent to

$$\min_{A, Z_k} \sum_k \psi_k^{-1} \|X_k - BY_k\|_2^2 \quad (4)$$

$$B \in R^{N \times M}, \quad Y_k \in R^{M \times L}$$

Where B is eigenvectors matrix of $\sum_k \psi_k^{-1} X_k X_k^{-1}$ with M largest eigenvalues, and $Y_k = (B^T B)^{-1} B^T X_k$.

Each iteration in MAP-CSP algorithm has two phases. First, determining a low dimensional subset of the signal in each class compared to the original signal. Then, finding jointly decelerated axes on each class data. By estimating A, the spatial filter can be represented as $W = [\hat{A}^T \hat{A}]^{-1} \hat{A}^T$.

After calculating the spatial filter (W), the projection signal (Z) from EEG sample (X) is given by: $Z = WX$. Generally, to create features, a small number of first and last m rows of Z are used by the following equation:

$$g_p = \log \left(\frac{\text{var}(Z_p)}{\sum_{i=1}^{2m} \text{var}(Z_i)} \right), \quad (5)$$

In order to create feature subset, MAP-CSP algorithm has been used by various numbers of M. In other words, MAP-CSP algorithm was evaluated for several times (k) and 2m feature vectors related to the particular M were extracted each time. As a result, we obtained 2mk features from each EEG sample to reach the following feature set:

$$g = \begin{bmatrix} f_{1,1} & f_{1,2} & \cdots & f_{1,2m} \\ f_{2,1} & \ddots & & f_{2,2m} \\ \vdots & & \ddots & \vdots \\ f_{N,1} & f_{N,2} & \cdots & f_{N,2m} \end{bmatrix}, \quad (6)$$

$$G = [g_1 \quad g_2 \quad \cdots \quad g_k] = \begin{bmatrix} f_{1,1,1} & \cdots & f_{1,2m,1} & f_{1,1,2} & \cdots & f_{1,2m,2} & \cdots & f_{1,2m,p} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ f_{N,1,1} & \cdots & f_{N,2m,1} & f_{N,1,2} & \cdots & f_{N,2m,2} & \cdots & f_{N,2m,p} \end{bmatrix}, \quad (7)$$

B. Sparse Feature Selection

Sparse regression models have been proposed so as to obtain significant features from feature set matrix. Sparse regression represents the data matrix as a linear combination of overcomplete components and is a successful approach to feature selection.

Here, the well-known lasso was used to estimate MAPCSP feature selection. Lasso algorithm transforms the sparsity to the convex issue and is superior to greedy algorithms. Lasso method minimizes the usual sum of squared errors with a bound on the sum of the absolute values of coefficients. Mathematical explanation of sparse feature selection can be presented as:

$$\beta = \arg \min_{\beta} \frac{1}{2} \|y - G\beta\|^2 + \lambda \|\beta\|_1, \quad (8)$$

Where y is the vector containing class labels {1,2}, and G is the feature set matrix with the ith column representing ith feature variable, β is the learned sparse coefficient vector and λ is the regularization parameter which controls sparsity of β . To evaluate sparse feature selection, we use SpaSM toolbox in MATLAB, a toolbox for sparse statistical modeling [18]. The column vectors in G, corresponding to those zero entries in β , are excluded to form an optimized feature set.

III. EXPERIMENT AND RESULTS

The performance of proposed method has been evaluated on the BCI competition IV, dataset 2a. This available data set is comprised of EEG data from 9 healthy subjects recorded using 22 channels in four-class motor imagery tasks, namely, left hand, right hand, foot, and tongue. Two sessions for each subject, one for training and the other for testing, were recorded. Each session contains 288 trials for 4 class labels.

A. Evaluation Scheme

In the evaluation phase, The EEG data from 0.5 to 2.5 seconds has been filtered by a band-pass filter (using 8-40 Hz fifth order Butterworth) after the visual cue. To create feature set, we evaluate MAP-CSP for 17 times (k) (from $M = 5$ to 22) and $m=3$ of the filters were used for each applied MAP-CSP. After the optimal MAP-CSP features were selected, the Linear Discriminant Analysis (LDA) was trained to classify MI tasks. Figure 4 Shows the validation based on regularization parameter (λ) in (8). And the results are based on the best λ which has the maximum accuracy. Some other methods like AIC and BIC could also be used for parameter estimation in Lasso algorithm, however, λ was chosen according to maximum accuracy.

According to the (8), as λ increases, the coefficient of β becomes exactly zero and if $\lambda=0$, the cost function just reforms to Mean Square Error (MSE). The role of the regularization parameter λ in choosing the discriminant features is shown in Figure 4. In this study, the optimized λ for each subject was determined by cross-validation on training data.

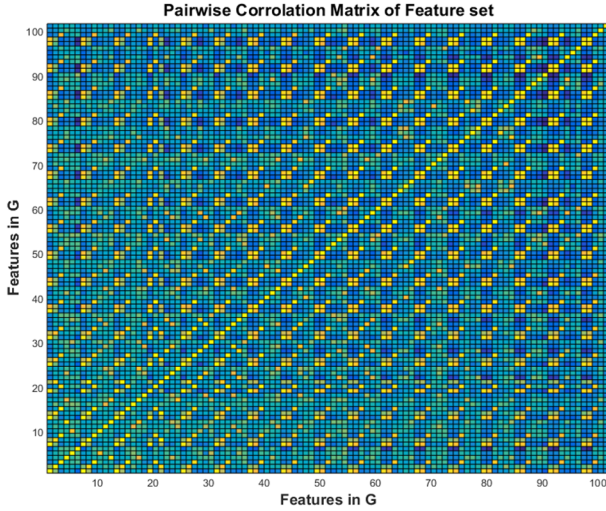


Figure 2. Pairwise correlation comparison for feature set. Bright elements have high correlation

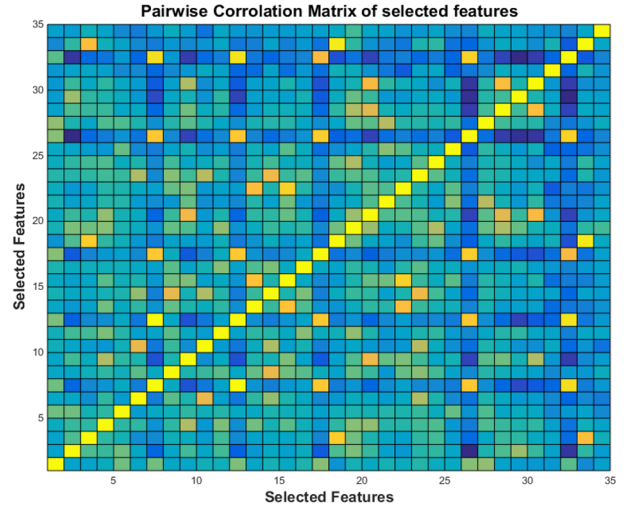


Figure 3. Pairwise Correlation comparison for selected features by sparse algorithm.

The main issue to be justified is whether, according to the role of M , extracted features of any M number could cover those features with smallest M . In other words, $2m$ features with $M = i$ may cover those features with $M = i-1$. To check and handle this issue, the pairwise correlation of G is represented in Figure 2.

As anticipated, Fig. 2 shows the correlation of features. The correlation of features is intensified by a period of $2m$, but the quantity and structure of correlation map is not equal in overall. In the next stage, sparse feature selection eliminates the correlation problem. Sparse dictionary learning has the property of evaluating dictionary with the minimum correlation of features. To endorse mentioned operation of sparse methods, correlation matrix has been shown in Fig. 3 after feature selection for 35 selected features. Here selected features have no correlation and are more discriminant from feature set.

B. Results

Classification accuracies derived by proposed method have been shown in Table. 1. In Table. 1, we compared classification accuracies of the filter bank MAP-CSP, MAP-CSP with the best M number obtained by sweeping, and automatic detection of M for each subject. As the results represent, the average of accuracy can be enhanced. These results show that the Sparse Filter Bank MAP-CSP improved the accuracies of MAP-CSP obtained by sweeping M and MAP-CSP (information-based source number estimation). Our proposed method resulted in higher classification accuracies for some subjects. Furthermore, in terms of computation time, finding the best parameter in MAP-CSP and Sparse Feature Selection is the most time-consuming part. Regarding this issue, the sparse feature selection with sparse methods has more efficient running time (averagely about 30 seconds less) in comparison to selection best M according to maximum accuracy.

Table. 1. Comparison of test classification accuracies for our proposed method and previous probabilistic methods on dataset 2a from BCI Competition IV

	Methods	Subjects									Mean
		#1	#2	#3	#4	#5	#6	#7	#8	#9	
Class 1 Left hand Vs. Right hand	Sparse FBMAP-CSP	90.97	61.11	99.30	73.61	69.44	70.83	77.77	96.52	94.44	81.55
	MAP-CSP (Automatic method)	90.27	58.33	95.13	68.05	59.72	63.8	70.8	94.44	93.75	77.14
	MAP-CSP (Best M)	92.36 (18)	60.41 (19)	97.22 (20)	97.22 (21)	65.97 (17)	70.83 (19)	78.47 (19)	97.91 (22)	94.44 (21)	83.87
Class 2 Left hand Vs. Foot	Sparse FBMAP-CSP	98.61	76.38	95.13	90.97	69.44	67.36	98.61	89.58	96.52	86.95
	MAP-CSP (Automatic method)	97.22	70.83	92.36	89.58	57.63	68.05	86.80	84.72	93.75	82.32
	MAP-CSP (Best M)	98.61 (18)	79.16 (19)	72.22 (11)	92.36 (10)	70.83 (13)	72.91 (20)	95.83 (9)	86.80 (6)	95.13 (6)	84.87

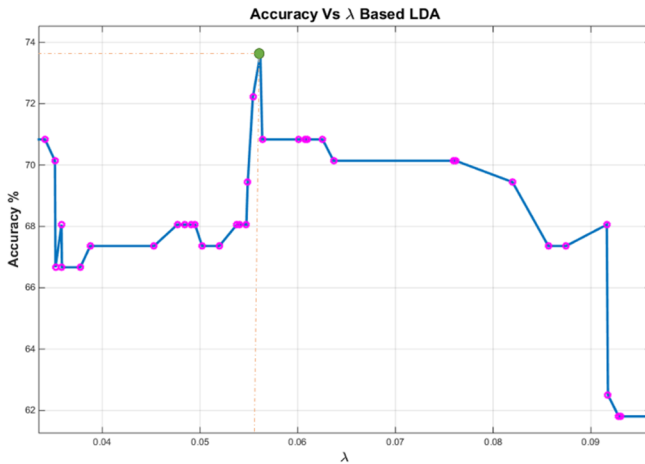


Figure 4. Effects of varying sparse regularization parameter λ on the test classification accuracies. The λ value with the maximum accuracy is highlighted in a green circle

Fig. 4 represents the variety of accuracy with respect to λ , for subject 4 in class 1, and the optimized λ was selected for the maximum accuracy. This procedure has been used to report maximum accuracy for all subjects in corresponding classes.

IV. CONCLUSION AND DISCUSSION

This paper proposed a novel method to collect more discriminant features from computed spatial filters by MAP-CSP. The Results from the BCI Competition IV dataset 2a revealed that the classification accuracy is increased by average about 2 percent in comparison with the using of MAP-CSP with the best number of independent source. Moreover, the results significantly yielded less running time, improved by 30 seconds in comparison with MAP-CSP. Evaluation of the mentioned method to select the optimized λ is time-consuming and not applicable to the small size feature set due to the limitation of Lasso algorithm. One of the effective algorithms for this issue is Sparse Bayesian Learning which estimates the model parameters based on evidence framework and has been used in recent researches on EEG signals.

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